MIL-D-003718B(EA)
31 October 1977
USED IN LIEU OF
MIL-D-3718A
15 July 1963

MILITARY SPECIFICATION

DYE MIX, DISPERSE RED 9-DEXTRIN

This limited coordination military specification has been prepared by US Army Armament Research and Development Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-D-3718A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers a mixture of Disperse Red 9 dye and dextrin.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
RR-S-366 - Sieve, Test.

FSC 6820

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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MILITARY

MIL-D-003284 - Dye, Disperse Red 9.
MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-D-3994 - Dextrin, Technical.

STANDARDS

FEDERAL

Fed. Std. No. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193 - Reagent Water.
D1895 - Apparent Density, Bulk Factor, and Pourability of Plastic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Composition. Dye Mix, Disperse Red 9-Dextrin shall be a thoroughly blended mixture of Dye, Disperse Red 9 conforming to MIL-D-003284 and Dextrin, Technical conforming to MIL-D-3994. No other materials shall be used in the preparation of the dye mix.

3.2 Chemical and physical characteristics. Dye Mix, Disperse Red 9-Dextrin shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement	Test paragraph
1-Methylaminoanthraquinone, percent by weight (dry basis)	85 ± 2	4.2.4.1
Dextrin, percent by weight, minimum	3.2	4.2.4.2
Volatile matter, percent by weight maximum	2.5	4.2.4.3
Particle size, percent by weight passing:		4.2.4.4
250-micron (No. 60) sieve, minimum	97	
149-micron (No. 100) sieve, minimum	90	
74-micron (No. 200) sieve, minimum	70	
Apparent density, grams per milliliter, minimum (dry basis)	0.35	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the Dye Mix, Disperse Red 9-Dextrin produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). No more than one lot of Disperse Red 9 dye shall be used to prepare a lot of dye mix.

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Three representative specimens of approximately 200 grams (g) each shall be randomly removed from the lot or batch of dye mix offered for acceptance and shall be placed in separate clean, dry containers labeled to identify the lot represented.

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled and closed shipping container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Polyethylene liner or closure thereof not as specified (level B only)
(d) Container closure not as specified
(e) Container damaged or leaking
(f) Desiccant not as specified or missing (level B only)
(g) Marking incorrect, missing, or illegible
(h) Palletization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used through out the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 1-Methylaminoanthraquinone. Prepare and measure the absorbance of a standard Disperse Red 9 dye solution (see 6.4) and a solution of the specimen as follows: Weigh to the nearest milligram (mg) 415 ± 5 mg of the standard dye or specimen which has been previously dried to a constant weight in an oven at 70° to 80° C and quantitatively transfer to a 500-milliliter (ml) beaker, using small quantities of benzene for washing. Stir to break up all lumps in order to insure complete wetting and to facilitate solution, heating if necessary. Transfer quantitatively to a 500-ml volumetric flask, dilute to the volume mark with benzene, and mix throughly. Filter through a 15-centimeter fluted filter paper and discard the first 300 ml of filtrate. Remove a 20-ml aliquot from the remaining 200 ml of filtrate, transfer to a second 500-ml volumetric flask, dilute to the volume mark with benzene, and mix thoroughly. Using a suitable spectrophotometer, measure the absorbance value of the more dilute solution at 500 ± 2 millimicrons. Use the benzene used in preparing the solution as the reference liquid in the spectrophotometer. Cell spacers or variable path length cells may be used in lieu of the prescribed dilution procedures. Calculate the percent by weight 1-methylaminoanthraquinone as follows:

$$\text{Percent 1-methylaminoanthraquinone} = \frac{ABC}{DE}$$

where: A = Percent purity of standard dye,
B = Absorbance value of specimen,
C = Weight of standard dye in grams,
D = Absorbance value of standard dye, and
E = Weight of specimen in grams.

4.2.4.2 Dextrin.

(a) Solution I. Dissolve 40 g of cupric sulfate ($CuSO_4.5H_2O$) in 1 liter of water.

(b) Solution II. Dissolve 200 g of potassium-sodium tartrate ($KNaC_4H_4O_6.4H_2O$) and 150 g of sodium hydroxide in water and dilute to 1 liter.

(c) Procedure. Weigh 0.500 g of specimen into a 250-ml Erlenmeyer flask. Add 100 ml of water and 5 ml of concentrated hydrochloric acid and heat until just before boiling begins. Remove the flask from heat and filter the solution through a filter crucible, rinsing with a small amount of hot water. Heat the solution to boiling and simmer for 3 hours adding water as necessary to maintain a volume of 100 ml. After this 3-hour period, continue boiling but allow the solution to evaporate to 20 ml. Remove from heat and allow to cool to room temperature. Add 20 ml of solution I and 40 ml of solution II and filter through a filter crucible to

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remove the material which precipitates from the basic solution. Heat the filtrate over a Bunsen burner which is regulated so that the solution begins to boil in about 4 minutes and continue to boil for exactly 3 minutes. (It is important that these directions be strictly observed.) Allow the red precipitate (Cu_2O) to settle and then filter through a filter crucible. Rinse the precipitate with ethanol and then dry for 1/2 hour at 105° ± 5° C. Weigh the precipitate and calculate the percent by weight dextrin as follows:

$$\text{Percent dextrin} = \frac{100A}{W}$$

where: A = Grams of dextrin equivalent to the grams of the precipitate (Cu_2O) as obtained from table II, and
W = Weight of specimen in grams.

4.2.4.3 Volatile matter. Weigh to the nearest milligram approximately 5 g of the specimen in a tared, glass-stoppered weighing bottle. Remove the stopper and dry to constant weight at 70° to 75° C. Cool to room temperature in a desiccator and weigh. Calculate the percent by weight volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100(A - B)}{W}$$

where: A = Weight of specimen and stoppered bottle before heating in grams,
B = Weight of specimen and stoppered bottle after heating in grams, and
W = Weight of specimen in grams.

4.2.4.4 Particle size. Use sieves conforming to RR-S-366. Nest the sieves in order of decreasing mesh size (with the sieve of largest mesh on top) on a receiving pan. Weigh to the nearest 0.1 g approximately 10 g of the specimen and then use one of the following procedures:

(a) Dry method. Place the weighed specimen on the top sieve and brush the material with a camel's-hair brush until no more specimen passes through the sieve. Remove the sieve and weigh the material retained. Repeat this procedure with each of the other sieves. Calculate the percent material passing through each sieve.

(b) Wet method. Mix the weighed specimen with sufficient water in a beaker by stirring with a glass rod to produce a smooth paste. Carefully wash the thoroughly wetted specimen through the sieves. (The use of a wetting agent is permitted.) Dry the sieves in an oven at 70° to 75° C and weigh the material remaining on each sieve. Calculate the percent material passing through the sieve.

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TABLE II. Dextrin equivalent to cuprous oxide

Cu_2O, mg	Dextrin, mg	Cu_2O, mg	Dextrin, mg	Cu_2O, mg	Dextrin, mg	Cu_2O, mg	Dextrin, mg
10	4.0	57	24.3	104	45.1	151	66.3
11	4.5	58	24.8	105	45.5	152	66.8
12	4.9	59	25.2	106	46.0	153	67.2
13	5.3	60	25.6	107	46.4	154	67.7
14	5.7	61	26.1	108	46.9	155	68.2
15	6.2	62	26.5	109	47.3	156	68.6
16	6.6	63	27.0	110	47.8	157	69.1
17	7.0	64	27.4	111	48.2	158	69.5
18	7.5	65	27.8	112	48.7	159	70.0
19	7.9	66	28.3	113	49.1	160	70.4
20	8.3	67	28.7	114	49.6	161	70.9
21	8.7	68	29.2	115	50.0	162	71.4
22	9.2	69	29.6	116	50.5	163	71.8
23	9.6	70	30.0	117	50.9	164	72.3
24	10.0	71	30.5	118	51.4	165	72.8
25	10.5	72	30.9	119	51.8	166	73.2
26	10.9	73	31.4	120	52.3	167	73.7
27	11.3	74	31.8	121	52.7	168	74.1
28	11.8	75	32.2	122	53.2	169	74.6
29	12.2	76	32.7	123	53.6	170	75.1
30	12.6	77	33.1	124	54.1	171	75.5
31	13.1	78	33.6	125	54.5	172	76.0
32	13.5	79	34.0	126	55.0	173	76.4
33	13.9	80	34.4	127	55.4	174	76.9
34	14.3	81	34.9	128	55.9	175	77.4
35	14.8	82	35.3	129	56.3	176	77.8
36	15.2	83	35.8	130	56.8	177	78.3
37	15.6	84	36.2	131	57.2	178	78.8
38	16.1	85	36.7	132	57.7	179	79.2
39	16.5	86	37.1	133	58.1	180	79.7
40	16.9	87	37.5	134	58.6	181	80.1
41	17.4	88	38.0	135	59.0	182	80.6
42	17.8	89	38.4	136	59.5	183	81.1
43	18.2	90	38.9	137	60.0	184	81.5
44	18.7	91	39.3	138	60.4	185	82.0
45	19.1	92	39.8	139	60.9	186	82.5
46	19.6	93	40.2	140	61.3	187	82.9
47	20.0	94	40.6	141	61.8	188	83.4
48	20.4	95	41.1	142	62.2	189	83.9
49	20.9	96	41.5	143	62.7	190	84.3
50	21.3	97	42.0	144	63.1	191	84.8
51	21.7	98	42.4	145	63.6	192	85.3
52	22.2	99	42.9	146	64.0	193	85.7
53	22.6	100	43.3	147	64.5	194	86.2
54	23.0	101	43.8	148	65.0	195	86.7
55	23.5	102	44.2	149	65.4	196	87.1
56	23.9	103	44.7	150	65.9	197	87.6

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4.2.4.5 Apparent density. Determine apparent density of the specimen in accordance with ASTM D1895, method A except that the specimen shall be dried as specified in 4.2.4.3 prior to testing.

5. PACKAGING

5.1 Packing. Dye Mix, Disperse Red 9-Dextrin shall be packed level B or commercial as specified (see 6.2).

5.1.1 Level B. Uniform quantities of no less than 68 and no more than 136 kilograms ± 1/4 percent of the marked net weight of dye mix shall be packed in a fiber drum conforming to rule 51, section 2 of the Uniform Freight Classification for a weight limit of over 225 but not over 300 pounds, with an aluminum barrier of 7.62 micrometers minimum thickness incorporated into one of the inner plies of the sidewall. Each drum shall be provided with a close fitting bag liner formed from polyethylene of 101.6 micrometers minimum thickness conforming to type I, class 1, grade and finish optional of L-P-378. Seams shall be completely heat sealed and shall meet the heatseal strength requirements of L-P-378. The filled bag shall be closed by heat sealing, tying, or knotting. A minimum of eight units of desiccant conforming to MIL-D-3464 shall be placed on top of the closed liner. Drum closure shall be fully gasketed and shall form a tight seal between the rim and the top of the drum.

5.1.2 Commercial. Uniform quantities of no less than 68 and no more than 136 kilograms ± 1/4 percent of the marked net weight of dye mix shall be packed in accordance with Fed. Std. No. 356.

5.2 Marking. Marking for level B shipments shall be in accordance with MIL-STD-129. Marking for commercial shipments shall be in accordance with Fed. Std. No. 356. All shipments shall be marked to show lot number and date of manufacture. In addition, each container shall be durably and legibly marked with contrasting letters and background to show the net weight of contents and the following information:

CAUTION

Avoid contact with skin or clothing.
In case of contact, flush with water.
Avoid breathing dust or fumes.
Use with adequate ventilation.

5.3 Palletization. Shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 except that the pallet shall be as specified in the contract (see 6.2).

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6. NOTES

6.1 Intended use. Dye Mix, Disperse Red 9-Dextrin is intended for use in the manufacture of red signaling smoke mixtures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.1).
(c) Type of pallet required (see 5.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufacutred by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Dye of known purity. 1-Methylaminoanthraquinone dye of known purity for use in 4.2.4.1 may be obtained from Commander, US Army Armament Research and Development Command, Product Assurance Directorate, Chemical Systems Division, Aberdeen Proving Ground, MD 21010. Alternatively, the contractor may prepare his own standard dye and obtain verification of its purity from Chemical Systems Division.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practice for Indicating Which Places of Figures Are To Be Considered Significant in Specified Limiting Values (ASTM E29).

Custodian:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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